

Vedant Palit

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EDUCATION

Indian Institute of Technology (IIT) Kharagpur, India

12/2021 – 04/2026

Btech in Industrial Engineering and Mtech in Financial Engineering

8.74/10

Relevant Coursework: Scalable Data Mining, Big Data Analysis, Regression and Time Series Models, Statistical Learning, Transform Calculus, Operations Research, Programming and Data Structures, Linear Algebra - Numerical and Complex Analysis

PUBLICATIONS

- How Do Linear Probes Emerge? A Circuit-Tracing Framework with Concept-Targeted Attribution - **Vedant Palit**, Florent Draye, Terry Jingchen Zhang, Bernhard Schölkopf, Zhijing Jin (*EMNLP 2026 Main Track*).
- Simulating Eutopia: Revisiting Long-term Fairness with Outcomes, Performativity, and Dynamics - **Vedant Palit**, Udvas Das, Brahim Driss, Debabrota Basu (*ECAF 2026 - Spotlight, ICML 2026 AI4LAW Workshop*).
- CLT-Forge: A Scalable Library for Cross-Layer Transcoders and Attribution Graphs - Florent Draye, **Vedant Palit**, Abir Harrasse, Tung-Yu Wu, Jiarui Liu, Punya Syon Pandey, Roderick Wu, Chih-Hao Hsu, Terry Jingchen Zhang, Zhijing Jin, Bernhard Schölkopf (*EMNLP 2026 Demo Track*).
- Towards Vision-Language Mechanistic Interpretability: A Causal Tracing Tool for BLIP - **Vedant Palit**, Rohan Pandey, Aryaman Arora, Paul Pu Liang (*ICCV 2023 CLVL Workshop*).
- Forgotten Polygons: Multimodal Large Language Models are Shape-Blind - William Rudman*, Michal Golovanevsky*, Amir Bar, **Vedant Palit**, Yann LeCun, Carsten Eickhoff, Ritambhara Singh (*ACL 2025 Findings Track*).
- What Do VLMs NOTICE? A Mechanistic Interpretability Pipeline for Gaussian-Noise-free Text-Image Corruption and Evaluation - Michal Golovanevsky*, William Rudman*, **Vedant Palit**, Ritambhara Singh, Carsten Eickhoff (*NAACL 2025*).
- WellDunn: On the Robustness and Explainability of Language Models and Large Language Models in Identifying Wellness Dimensions - Seyedali Mohammadi, Edward Raff, Jinendra Malekar, **Vedant Palit**, Francis Ferraro, Manas Gaur (*EMNLP 2024 Blackbox NLP Workshop*).
- Knowledge Graph Guided Semantic Evaluation of Language Models For User Trust - Kaushik Roy, Tarun Garg, **Vedant Palit** (*IEEE CAI 2023*).
- Exploring The Potential of Large Language Models for Assisting with Mental Health Diagnostic Assessments - Kaushik Roy, Harshul Surana, Darssan Eswaramoorthi, Yuxin Zi, **Vedant Palit**, Ritvik Garimella, Amit Sheth (*ACM Transactions on Computing for Healthcare*).

RESEARCH EXPERIENCE

Research Lead | Eurosafe AI | Jinesis AI Lab | MPI-IS

Tübingen, Germany

Advised by Dr Zhijing Jin and Dr Bernhard Schölkopf

Jan 2026 – Present (On-site from July)

- Introduced Concept-Targeted Attribution , redirecting attribution graphs toward linear probe directions to mechanistically explain internal concept representations, demonstrating dissociation between concept-encoding and token-generation circuits and predicting probe accuracy from graph topology across Gemma-2-2B and Llama-3.2-1B.
- Devised a compression and quantization framework for the efficient saving and loading of activations from the GPT-2 and Llama-1B model for the open-source training of Cross-Layer Transcoders.

Research Collaborator | University of Tübingen | Brown University Providence, RI (Remote)

Advised by Dr Ritambhara Singh and Dr Carsten Eickhoff

Feb 2024 – Feb 2026

- Analysed the effectiveness of a novel semantic minimal pair and symmetric text replacement corruption scheme, against Gaussian noise-based causal mediation analysis through activation patching and knockouts.
- Evaluated the multi-step math reasoning capability for the LLaVA-1.5, LLaVA-Next, LLaVA-OneVision, Qwen2-VL, Molmo, InternVL, LLaMA-3.2 and Math-PUMA vision-language models.

Research Intern | Inria, University of Lille
Advised by Dr Debabrata Basu

Lille, France (Remote)
Dec 2023 – May 2026

- Proposed and developed a standardised gymnasium environment for the evaluation of algorithmic fairness in the approval of loans through credit worthiness.
- Benchmarked standard RL algorithms such as Linear-Q, Discrete-Q, Proximal Policy Gradient and performative RL algorithms across social welfare, rawlsian, fairness aware and utilitarian reward functions.

Research Collaborator | Carnegie Mellon University
Advised by Rohan Pandey, Aryaman Arora and Dr Paul Pu Liang

Pittsburgh, PA (Remote)
April 2023 – Dec 2023

- Proposed and developed a pipeline adapting causal mediation analysis utilising Gaussian Noise injection and activation patching to interpret blackbox architectures of VL transformers.
- Demonstrated the methodology on BLIP and used the COCO-QA dataset to study the internal layer effect of the multimodal text-encoder on model outputs as well as analysed the variation of patching effects with the strength of the noise.

Research Intern | University of Maryland, Baltimore
Advised by Dr Manas Gaur

Baltimore, MD (Remote)
Nov 2022 – Sep 2024

- Trained various general and domain-specific models for suicide risk assessment, using the cross entropy and gamblers loss functions on annotated datasets containing social media posts classified into 6 different wellness dimensions.
- Utilised singular value decomposition to analyse the impact of the loss function on the attention scores of the models.

INDUSTRY EXPERIENCE

Machine Learning Intern | JP Morgan Chase
Model Risk Governance and Review Division

Bangalore, India
May 2025 – July 2025

- Investigated the internal mechanisms of a LLaMA-3.1 model fine-tuned on confidential JP Morgan data, utilising causal mediation analysis and activation patching.
- Reviewed quantitative models through backtesting and classical interpretability techniques like SHAP and Partial Dependence Plots.

RESEARCH ALIGNED ACTIVITIES

Conference Reviewing: Reviewed for ICLR 2025, NeurIPS Mech Interp Workshop 2025, ICLR 2026 and NeurIPS 2026.

Mentoring: Guided over 25 students on research directions, career planning, and academic development.

Technical Writing: Authored blogs reviewing papers on ML, DL, and AI. [Medium]

Research Communities: Member of the Association for Computing Machinery (ACM).

TECHNICAL SKILLS

Programming Languages: C/C++, Python, MATLAB
CV-NLP: Transformers, OpenCV, PIL, Llama-Index

ML-DL: TensorFlow, Pytorch, Torchvision, Sklearn, Caffe
Miscellaneous: Mysql, LaTeX, HTML, Markdown, Git

AWARDS AND ACHIEVEMENTS

Master’s Thesis Project: Ranked in the top 10% of the dept with the highest grade in both components of the MTP.

Adobe AI Challenge: Captained the team of IIT KGP to Gold among 23 competing IITs at Inter IIT Tech Meet 2024.

JEE Advanced: Placed in the top 1.0% nationally among candidates appearing in JEE Advanced, 2021.

JEE Mains: Placed in the top 0.8% nationally among candidates appearing in JEE MAIN 2021.

WBJEE: Placed in the top 0.1% in the state among candidates appearing in WBJEE 2021

Scientific Forum: Selected as a delegate to represent India at the Asia Pacific Forum for Science Talented 2019.

EXTRACURRICULARS

NSS Volunteer: Recipient of the gold medal for exceptional service work as an active participant in cleanliness drives, clothes distribution drives and education camps conducted by the NSS in villages near Kharagpur.

Blogs on history: Writer of a series of blogs on history and prehistory and their implications on the world. [Blog]

College Societies: An active member of the IIT Kharagpur Quiz Club.